QUIZ 9-A (Spring-2024)

ID:____________ **NAME:_________________________________** **SECTION: ____**

1. Design the sequential circuit using **JK-Flip-flop** following given circular sequence **2->3->5->7** back to 2? find self-correcting state **0,1** and force correction for state **4,6** to state **2** for unused state? You are required to produce state table and state equations.
   Self-Correction state 0 is ______ and **state 1**_____

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2. For the following table, tabulate **reduce state table**. Start from state a, and input sequence **01110011**, determine output sequence.

	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

3. Design **4-bit-Universal Shift Register**? Perform following operations.

S_2	S_1	S_0	Register Operation	S_2	S_1	S_0	Register Operation
0	0	0	Clear to 0	1	0	0	Compliment Of Old State
0	0	1	Parallel load	1	0	1	Shift Left
0	1	0	Shift Right	1	1	0	Compliment of Parallel load
0	1	1	No Change	1	1	1	Clear to 1